

ALEXANDER GILBREATH

MACHINE LEARNING · COMPUTER VISION · SOFTWARE ENGINEERING

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Artificial Intelligence undergraduate at UC San Diego (junior standing, GPA 3.6) who builds and validates machine-learning systems end to end — a GPT-style transformer built from scratch, a shipped AI-vision iOS app, and a disaster-damage vision transformer. Strong software-engineering foundation in Python and C/C++ spanning deep learning, computer vision, and model-accuracy validation. Winner of the 2025 OpenArt AI Hackathon, with a direct technical reference from a CSE professor. Seeking a machine-learning or software-engineering internship; authorized to work in the U.S. without sponsorship.

EDUCATION

University of California, San Diego — B.S. Artificial Intelligence *Junior standing*

Expected June 2029 · GPA 3.6

Relevant coursework: Introduction to AI · Systems Programming in C (CSE 29) · Computer Organization & Assembly (CSE 30, A) · Discrete Mathematics (CSE 20, A) · Data Structures (CSE 12)

TECHNICAL SKILLS

AI & ML	PyTorch, deep learning, transformer architecture (built from scratch), LLM APIs and prompting, computer vision, vision transformers, CUDA-accelerated GPU training, model evaluation and accuracy validation
Programming	Python, C, C++, Java, TypeScript, ARM assembly
Tools	Git, GitHub, Cursor, Linux, Jupyter, gcc, gdb, valgrind, Make, React, Vite
Testing	Ground-truth dataset construction, accuracy benchmarking, unit testing, edge-case documentation

PROJECTS

GPT Language Model from Scratch *PyTorch, Python* 2026

- Built and trained a 30M-parameter decoder-only transformer from scratch, implementing attention, tokenization, and the full training loop end to end.
- Drove training loss from 6.11 to 3.87 through iterative hyperparameter tuning and GPU-accelerated training.

CompGym — AI Nutrition & Training Log *iOS, AI Vision* 2026

- Contributed from requirements through release testing of an iOS app using AI vision to identify food and estimate macronutrients from a single photograph.
- Built the accuracy-validation pipeline: constructed hand-measured ground-truth datasets and benchmarked model predictions to improve food identification and portion estimation.

Disaster-Damage Classifier *PyTorch, CUDA* 2025

- Trained a vision transformer on CUDA-accelerated GPUs to classify disaster-damage severity, collaborating on a team to tune hyperparameters iteratively for accuracy.

AI Accessibility Research *UC San Diego, advisor Sina Radi* 2025 – Present

- Contribute to applied computer-vision research developing AI communication tools for a nonverbal user with severe autism, from requirements through model validation.

Systems & Software Engineering *C, CSE 29 / 30*

- Built a malloc/free shared library with a clean public API (`vmLib.h`), a Unix shell using `fork/execvp/wait`, and a linked-list calendar — validated with assert-based unit tests and leak-free under `valgrind`.

EXPERIENCE & LEADERSHIP

UC San Diego Environmental Health & Safety *Course Auditor* 2025 – Present

- Audit AI-generated training content for accuracy, documenting model output errors and hallucinations before campus-wide deployment; train Resident Assistants on corrected materials.

Engineering Innovators Entrepreneurship Club *President* 2025 – Present

- Elected Financial Chair, then Co-President, then President; founded and ran a 40-student hackathon with a \$1,000 prize pool, managing sponsors, judging, and operations end-to-end.

RedCoin Capital *Research Intern* 2025

- Studied subnet architectures and decentralized-AI infrastructure at a leading decentralized-AI firm (Kraken partner), translating research into technical briefs.

Tritons Thrift *Founder* 2022 – Present

- Founded a sustainable resale venture generating approximately \$5,000 in five months; scaling into a UCSD Associated Students retail partnership.

AWARDS & REFERENCES

Winner, 2025 OpenArt AI Hackathon *UC San Diego, presented by CSES*

References: Prof. Adalbert Gerald Soosai Raj (CSE 29, CSE 8A) · Prof. Barry Brown (CSE 21) — full contact details available on request.